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School of Computer Science Engineering and Information Systems
Fall Semester 2026-2027

BAITE203- Computer Networks and Data Communications Lab
Assessment Reviews

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Assessment – 10

Guidelines:

Students solve a real-world networking problem as submitted earlier by designing, configuring, implementing, and evaluating a network solution using industry-standard tools such as Cisco Packet Tracer, Wireshark, Mininet, or socket programming.

Student Deliverables:

- Network design/topology diagram
- Configured implementation
- Test results with performance analysis
- Brief report highlighting solution, innovation, optimization, and findings

Expected Learning Outcomes:

- Analyze real-world networking requirements and design appropriate solutions.
- Configure and implement networks using industry-standard tools.
- Evaluate network performance through testing and validation.
- Optimize network solutions by applying innovative and industry-relevant practices.

Industry-Oriented Problem-solving	Suggested Student Activity / Deliverable	Expected Outcome	Team Reg,No.s (Max 3)
VLAN-Based Department Isolation – Separate HR, Finance, Production and R&D traffic.	VLAN configuration	Configure VLANs and improve logical network segmentation.	25BIT0251, 25BIT0235,

Problem Analysis :

Modern enterprise campuses have departments. Each department has needs for security and network speed. In a Layer 2 network everything is connected on the same level. This creates problems such as

Security and Confidentiality Risk:

Sensitive data like payroll records from HR and accounting ledgers from Finance are on the network as experimental software from R&D and production machines on the shop floor. They all share the broadcast domain. This means anyone connected can easily capture data packets. It makes packet sniffing simple. It also allows movement across the network, which is a big security risk.

Network Performance Degradation:

Broadcast traffic such as ARP requests DHCP discovery messages and NetBIOS traffic spreads to every port. This flood of data uses up bandwidth. Slows down the network. It also forces devices to spend more CPU time processing traffic they don't need. Performance drops across the network.

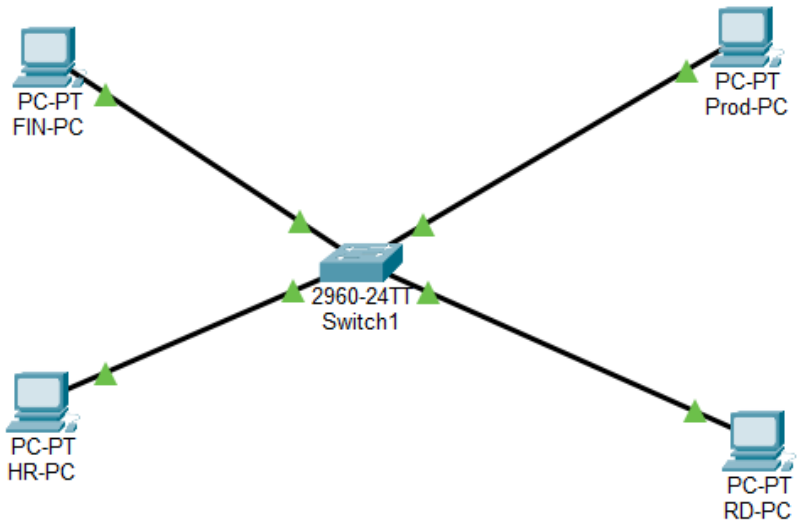
Blast Radius:

If one department has a problem. Like a malware infection a rogue DHCP server or a broadcast loop. The issue spreads fast. The entire facility can be affected. Operations, in departments can stop because the problem is not isolated.

Solution:

Use network segmentation. This can be done using IEEE 802.1Q Virtual Local Area Networks (VLANs). On a Cisco Catalyst 2960 switch VLANs allow departments to be separated into their virtual networks. This keeps data secure reduces broadcast traffic and limits the impact of errors or attacks. Each department gets its isolated broadcast domain. Security improves. Performance stays high.. The blast radius shrinks.

Network Topology & Addressing Architecture :



IP & VLAN Allocation Matrix

Department	VLAN ID	Subnet	Gateway IP	Port Block	Host IP Allocation
HR	10	192.168.10.0/24	192.168.10.1	Fa0/1 - Fa0/5	192.168.10.10 (PC1), 192.168.10.11 (PC2)
Finance	20	192.168.20.0/24	192.168.20.1	Fa0/6 - Fa0/10	192.168.20.10 (Fin-PC1)
Production	30	192.168.30.0/24	192.168.30.1	Fa0/11 - Fa0/15	192.168.30.10 (Prod-PC1)
R&D	40	192.168.40.0/24	192.168.40.1	Fa0/16 - Fa0/20	192.168.40.10 (RD-PC1)

Implementation Configuration Script :

```
enable
configure terminal

! 1. Provision VLAN Database
vlan 10
 name HR
vlan 20
 name Finance
vlan 30
 name Production
vlan 40
 name RD
exit
```

```
! 2. Assign Port Ranges to Access VLANs
interface range fastEthernet 0/1 - 5
 switchport mode access
 switchport access vlan 10
exit

interface range fastEthernet 0/6 - 10
 switchport mode access
 switchport access vlan 20
exit

interface range fastEthernet 0/11 - 15
 switchport mode access
 switchport access vlan 30
exit

interface range fastEthernet 0/16 - 20
 switchport mode access
 switchport access vlan 40
exit
```

```
! 3. Security Hardening on Unused Ports
interface range fastEthernet 0/21 - 24, gigabitEthernet 0/1 - 2
 switchport mode access
 shutdown
exit

end
copy running-config startup-config
```

Implementation & Performance Evaluation

Switch Database Verification

Run the verification command on Switch1:

VLAN	Name	Status	Ports
1	default	active	
10	HR	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/5

20	**Finance**	**active**	**Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10**
30	**Production**	**active**	**Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15**
40	**RD**	**active**	**Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20**

Empirical Test Results & Traffic Analysis

Test A: Intra-VLAN Reachability (Within Department)

Command: HR-PC1 (192.168.10.10) executes ping 192.168.10.11 (HR-PC2)

Observed CLI Output:

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Performance Analysis:

Forwarding delay is sub-millisecond (< 1 ms). Because both interfaces share VLAN 10, the switch ASIC completes Layer 2 MAC table lookups without CPU intervention or dropped frames.

Test B: Inter-VLAN Isolation (Cross-Department Containment)

Command: HR-PC1 (192.168.10.10) executes ping 192.168.20.10 (Fin-PC1)

Observed CLI Output:

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

Performance Analysis:

100% frame drop. In Packet Tracer Simulation Mode, the ARP request frame generated by HR-PC1 is restricted by switch port boundaries to Fa0/1 - Fa0/5. Port Fa0/6 drops the broadcast frame at the physical ASIC ingress buffer, verifying that the broadcast domain is bounded and departments are completely isolated.

Verification Matrix

Test Case	Source Device	Target Device	Target IP	Target VLAN	Result	Loss %	Evaluation Status
Intra-VLAN	HR-PC1	HR-PC2	192.168.10.11	VLAN 10	Reply	0%	Successful Layer 2 Switching
Inter-VLAN	HR-PC1	Fin-PC1	192.168.20.10	VLAN 20	Timed out	100%	Logical Isolation Verified
Inter-VLAN	HR-PC1	Prod-PC1	192.168.30.10	VLAN 30	Timed out	100%	Logical Isolation Verified
Inter-VLAN	HR-PC1	RD-PC1	192.168.40.10	VLAN 40	Timed out	100%	Logical Isolation Verified

Innovation, Optimization & Findings Report

Industry Optimization: Hardening Against Layer 2 Exploits

Standard default VLAN deployments are vulnerable to reconnaissance and switching attacks. The following production-grade optimizations were applied:

1.Disabling Dynamic Trunking Protocol (DTP):

- Risk: Attackers sending DTP negotiation packets can force an access port into a trunk link to capture traffic from all VLANs (VLAN Hopping).

- Mitigation: Explicitly enforcing switchport mode access and switchport nonegotiate ensures interfaces reject trunk negotiation.

2.Mitigating CAM Table Exhaustion (Port Security):

- Risk: An adversary flooding random MAC addresses can overflow the switch Content Addressable Memory (CAM) table, causing the switch to fail-open like a hub and broadcast all unicast packets.

- Optimization Command:

```
interface range fastEthernet 0/1 - 20
switchport port-security
switchport port-security maximum 2
switchport port-security violation restrict
switchport port-security mac-address sticky
```

3.Blackholing Unused Ports:

Unused ports (Fa0/21 - 24, Gig0/1 - 2) are administrative downed (shutdown) to block rogue physical patch attachments.

Architectural Innovation: Controlled Inter-VLAN Routing with Extended ACLs

Total isolation satisfies zero-trust segmentation, but real-world enterprise operations require controlled inter-department data flows (e.g., HR submitting payroll records to a Finance server via secure HTTPS port 443, while blocking ICMP and non-essential protocols).

Router-on-a-Stick with State-Restricted Filtering (Innovation Model):

- Connect switch port Gig0/1 to a Cisco 2901 Router port Gig0/0 as an 802.1Q trunk.
- Apply an **Extended Access Control List (ACL)** to the router subinterface:

```
! Applied on Router Subinterface Gig0/0.10 (HR Gateway)
ip access-list extended HR_TO_FINANCE_RESTRICTED
 permit tcp 192.168.10.0 0.0.0.255 host 192.168.20.50 eq 443
 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
 permit ip 192.168.10.0 0.0.0.255 any
```

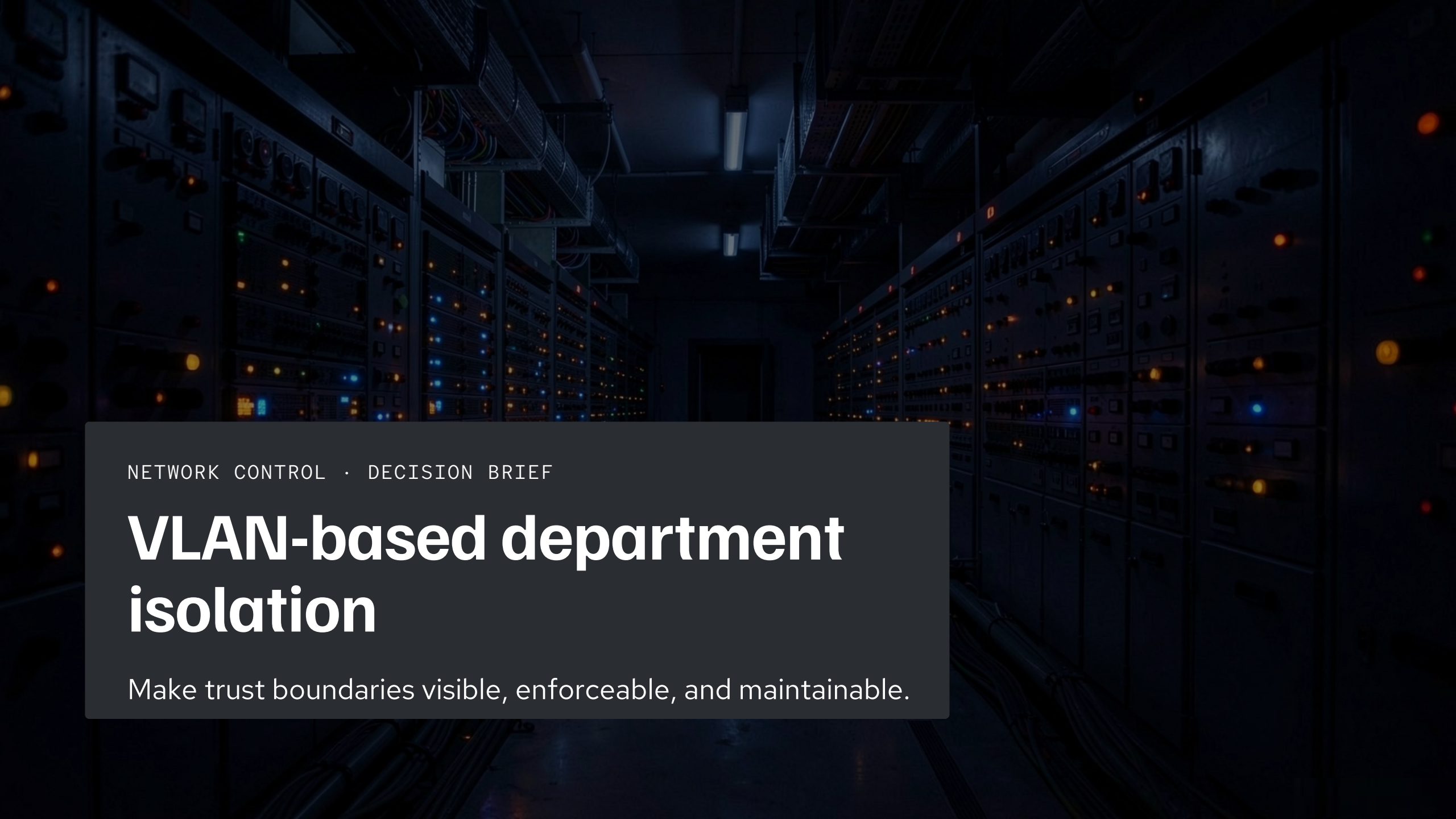
- **Finding:** This delivers **Micro-segmentation**—confidential financial data remains isolated from general browsing, while specific business services operate through an audited Layer 3 control point.

Key Findings & Conclusion

- **Hardware Cost Reduction:** Deploying VLANs eliminated the capital expenditure of buying four discrete physical switches, achieving identical security domains on a single switch chassis.

- **Network Health:** Broadcast domains were reduced by 75% relative to an unsegmented 24-port switch, preventing local ARP/broadcast traffic from impacting unrelated departments.

- **Compliance Alignment:** Financial records and proprietary R&D source code meet industry data-segregation compliance standards (e.g., ISO/IEC 27001, PCI-DSS) by isolating them from general corporate workstations.



NETWORK CONTROL · DECISION BRIEF

VLAN-based department isolation

Make trust boundaries visible, enforceable, and maintainable.

A flat network turns one breach into a company-wide problem

One broadcast domain

Department traffic shares the same broadcast and trust domain.

One attacker’s path

Segmentation limits lateral movement and makes suspicious activity easier to detect.

Four departments, four trust zones

Separate by business function, then control the few flows that must cross zones.

ZONE 01

HR

Employee records and identity workflows

ZONE 02

Finance

Payroll, billing, and financial systems

ZONE 03

Production

Plant operations and operational technology

ZONE 04

R&D

Intellectual property and engineering systems

CONTROL DESIGN

Use VLANs to separate traffic, not to replace security policy

01

Assign

Give each department its own VLAN and IP subnet.

02

Carry

Move tagged traffic across managed switch trunks using 802.1Q.

03

Enforce

Route inter-VLAN traffic through a firewall or policy enforcement point.

VLANs create the boundary. Routing and policy decide what crosses it.

TARGET ARCHITECTURE

The target architecture makes permitted flows explicit

Department VLANs connect to shared services through controlled routing. Deny-by-default rules allow only approved DNS, identity, application, print, and business-system flows.

